EXPERIMENT - 1

AIM:

Study of Various Network commands used in Linux and Windows

Basic NETWORKING COMMANDS:

arp-a:

ARP is short form of address resolution protocol. It will show the IP address of your Computer along with the IP address and MAC address of your router.

host name:

This is the simplest of all TCP/IP Commands. It simply displays the name of your Computer

ipConfig/all:

This command displays detailed Configuration information about your TCP/IP Connection including Router, Gateway, DNS, DHCP, and type of Ethernet adapter

nbtstat-a:

This command helps solve problems with NetBIOS name resolution It stands for NetBios over TCP/IP

Netstat:

It displays a variety of statistics about a Computers active TCP/IP Connections. It is a Command line tool for monitoring network Connections both incoming and outgoing

e.g:- netstat -r

nslookup:

It is a tool used to perform DNS lookups in Linux. It is used to Display DNS details, mx records for a domain. It can operate in two modes: interactive and non interactive eg: nslookup www.google.com.

pathping:

It is unique to Windows and is basically a Combination of the Ping and Tracert Commands.

Ping:

It is the best way to test connectivity between two

```
C:\Users\REC>arp -a

Interface: 172.16.8.56 --- 0x9
  Internet Address      Physical Address      Type
  172.16.8.1            7c-5a-1c-cf-be-45     dynamic
  172.16.8.7            d4-3d-7e-ad-3e-bf     dynamic
  172.16.8.48           f8-bc-12-69-80-01     dynamic
  172.16.8.62           f8-bc-12-8d-ed-be     dynamic
  172.16.10.66          d8-bb-c1-c5-6a-9d     dynamic
  172.16.10.84          d8-bb-c1-c5-c8-99     dynamic
  172.16.11.255         ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.102.18        01-00-5e-7f-66-12     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
```

```
C:\Users\REC>hostname
DESKTOP-ECC1JEK
```

```
C:\Users\REC>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::d988:bcf0:4035:27e8%9
   IPv4 Address. . . . . . . . . . . : 172.16.8.56
   Subnet Mask . . . . . . . . . . . : 255.255.252.0
   Default Gateway . . . . . . . . . : 172.16.8.1
```

```
C:\Users\REC>nbtstat -r

    NetBIOS Names Resolution and Registration Statistics
    ----------------------------------------------------

    Resolved By Broadcast     = 0
    Resolved By Name Server   = 0

    Registered By Broadcast   = 54
    Registered By Name Server = 0
```

```
C:\Users\REC>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    172.16.8.56:63464      sf-in-f188:5228        ESTABLISHED
  TCP    172.16.8.56:63602      104.18.32.47:https     ESTABLISHED
  TCP    172.16.8.56:63627      a23-44-10-56:https     CLOSE_WAIT
  TCP    172.16.8.56:63637      hkg12s32-in-f3:https   ESTABLISHED
  TCP    172.16.8.56:63639      ym-in-f94:https        ESTABLISHED
  TCP    172.16.8.56:63640      4.213.25.240:https     ESTABLISHED
  TCP    [fe80::d988:bcf0:4035:27e8%9]:1521  DESKTOP-ECC1JEK:49674  ESTABLISHED
  TCP    [fe80::d988:bcf0:4035:27e8%9]:49674  DESKTOP-ECC1JEK:1521   ESTABLISHED
```

```
C:\Users\REC>nslookup google.com
Server:  UnKnown
Address:  172.16.8.1

Non-authoritative answer:
Name:    google.com
Addresses:  2404:6800:4007:808::200e
          142.250.66.14
```

nodes

1. # ping hostname (ping localhost)
2. # ping ip address (ping 4.2.2.2)
3. # ping fully qualified domain name (ping.www.facebook.com)

Route:

route command is used to show the IP routing table. It is primarily used to setup static routes to specific host or networks via an interface.

Some important linux networking commands:

1. ip:

It is one of the basic commands every administrator will need daily work from setting up new systems and assigning IPs to troubleshooting existing systems

Syntax: ip <options> <OBJECT> <COMMAND>

eg: [root@server ~]# ip address show.

2. ifconfig:

It is a stable in many sysadmin's tool belt for configuring and troubleshooting networks.

3. mtr:

It is a program with a command-line interface that serves as a network diagnostic and troubleshooting tool. It combines the functionality of the ping and traceroute commands

syntax: mtr <options> hostname/IP

eg: [root@server ~]# mtr.google.com.

4. tcpdump:

This command is designed for capturing and displaying packets. you can install tcpdump with the command below:

[root@server ~]# dnf install -y tcpdump

5. Ping:

It is a tool that verifies IP-level connectivity to another TCP/IP computer by sending Internet Control Message Protocol Echo Request messages. The receipt of corresponding Echo Reply message is displayed along with round-trip times.

```
C:\Users\REC>pathping google.com

Tracing route to google.com [172.217.24.14]
over a maximum of 30 hops:
  0  DESKTOP-ECC1JEK [172.16.8.56]
  1  172.16.8.1
  2  137.97.163.129
  3  *        *        *
Computing statistics for 50 seconds...
            Source to Here   This Node/Link
Hop  RTT    Lost/Sent = Pct  Lost/Sent = Pct  Address
  0                                           DESKTOP-ECC1JEK [172.16.8.56]
                                 0/ 100 =  0%   |
  1    0ms     0/ 100 =  0%     0/ 100 =  0%  172.16.8.1
                                 0/ 100 =  0%   |
  2    2ms     0/ 100 =  0%     0/ 100 =  0%  137.97.163.129

Trace complete.
```

```
C:\Users\REC>route print
===========================================================================
Interface List
  9...00 27 0e 13 eb f8 ......Realtek PCIe GbE Family Controller
  1...........................Software Loopback Interface 1
===========================================================================

IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0      172.16.8.1      172.16.8.56     291
        127.0.0.0        255.0.0.0         On-link         127.0.0.1     331
        127.0.0.1  255.255.255.255         On-link         127.0.0.1     331
  127.255.255.255  255.255.255.255         On-link         127.0.0.1     331
      172.16.8.0    255.255.252.0         On-link      172.16.8.56     291
     172.16.8.56  255.255.255.255         On-link      172.16.8.56     291
    172.16.11.255  255.255.255.255         On-link      172.16.8.56     291
        224.0.0.0        240.0.0.0         On-link         127.0.0.1     331
        224.0.0.0        240.0.0.0         On-link      172.16.8.56     291
  255.255.255.255  255.255.255.255         On-link         127.0.0.1     331
  255.255.255.255  255.255.255.255         On-link      172.16.8.56     291
===========================================================================
Persistent Routes:
  Network Address          Netmask  Gateway Address  Metric
          0.0.0.0          0.0.0.0      172.16.8.1  Default
===========================================================================

IPv6 Route Table
===========================================================================
Active Routes:
 If Metric Network Destination      Gateway
  1    331 ::1/128                  On-link
  9    291 fe80::/64                On-link
  9    291 fe80::d988:bcf0:4035:27e8/128
                                    On-link
  1    331 ff00::/8                 On-link
  9    291 ff00::/8                 On-link
===========================================================================
Persistent Routes:
  None
```

```
C:\Users\REC>ping google.com

Pinging google.com [142.250.205.142] with 32 bytes of data:
Reply from 142.250.205.142: bytes=32 time=4ms TTL=118
Reply from 142.250.205.142: bytes=32 time=3ms TTL=118
Reply from 142.250.205.142: bytes=32 time=2ms TTL=118
Reply from 142.250.205.142: bytes=32 time=3ms TTL=118

Ping statistics for 142.250.205.142:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 4ms, Average = 3ms
```

```
[root@localhost ~]# dnf install -y tcpdump
CentOS Stream 9 - BaseOS                        1.7 kB/s | 5.3 kB     00:03
CentOS Stream 9 - AppStream                     1.5 kB/s | 5.4 kB     00:03
CentOS Stream 9 - AppStream                     3.3 MB/s |  28 MB     00:08
CentOS Stream 9 - Extras packages                12 kB/s | 5.7 kB     00:00
CentOS Stream 9 - Extras packages               4.7 kB/s |  21 kB     00:04
Package tcpdump-14:4.99.0-9.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
```

```
[root@localhost ~]# tcpdump -D
1.enp0s3 [Up, Running, Connected]
2.any (Pseudo-device that captures on all interfaces) [Up, Running]
3.lo [Up, Running, Loopback]
4.bluetooth-monitor (Bluetooth Linux Monitor) [Wireless]
5.usbmon2 (Raw USB traffic, bus number 2)
6.usbmon1 (Raw USB traffic, bus number 1)
7.usbmon0 (Raw USB traffic, all USB buses) [none]
8.nflog (Linux netfilter log (NFLOG) interface) [none]
9.nfqueue (Linux netfilter queue (NFQUEUE) interface) [none]
```

[root@server ~]# ping google.com.
PING google.com (216.58.206.174) 56(84) bytes of data
64 bytes from sof02s27-in-f14.1e100.net (216.58.206.174): icmp_
seq=1 ttl=56 time=10.7 ms

nmcli Command

1. nmcli Connection show
   Displays all available network Manager Connection profiles and their associated devices.
2. nmcli connection add con-name <Connection name> ifname <device-name> type ethernet
   Creates a new Ethernet Connection profile for the specified network interface.
3. nmcli Connection modify "wired Connection 1" Connection id Internal-LAN 2 modifies the settings of an existing network Connection
4. modify "Internet-LAN" ipv4 method auto configures the connection to automatically obtain an IPV4 address using
5. nmcli connection modify "Internal-LAN ipv4.method manual ipv4.address 192.0.2.1/24 ipv4-gateway 192.02.254 ipv4.dns 192.0.2.200.
   Assigns a static ipv4 address, gateway DNS Server, and DNS search Domain to the Connection.
6. nmcli connection up Internal-LAN ipv6.method auto
   Configures the Connection to automatically obtain an IPV6 address using SLAAC
7. nmcli Connection up Internal-LAN Activates the specified network Connection profile
8. nmcli Connection modify Internal-LAN ipv6.method manual ipv6.address 2001:db8:0::100164 ipv6 gateway 2001:2db8:1::1 ipv6-dns 2001:4860:4860:8888
   Assigns a static ipv6 address gateway DNS server to the Connection

```
ipv6.method:                            auto
```

```
[root@localhost ~]# nmcli connection modify  Internal-LAN ipv6.method manual ipv
6.addresses 2001:db8:1::100/64 ipv6.gateway 2001:db8:1::1 ipv6.dns 2001:4860:486
0::8888
```

```
ipv6.method:                            manual
ipv6.dns:                               2001:4860:4860::8888
ipv6.dns-search:                        --
ipv6.dns-options:                       --
ipv6.dns-priority:                      0
ipv6.addresses:                         2001:db8:1::100/64
ipv6.gateway:                           2001:db8:1::1
```

```
[root@localhost ~]# nmcli connection up Internal-LAN
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkMa
nager/ActiveConnection/3)
```

```
[root@localhost ~]# systemctl restart NetworkManager
```

```
[root@localhost ~]# nmcli device status
DEVICE  TYPE      STATE                   CONNECTION
enp0s3  ethernet  connected               Internal-LAN
```

```
[root@localhost ~]# nmcli connection show
NAME    UUID                                  TYPE      DEVICE
enp0s3  1cd79c34-f650-3d92-ae10-8f9710d22b92  ethernet  enp0s3
```

```
[root@localhost ~]# nmcli connection add con-name Internal-LAN ifname enp0s3 typ
e ethernet
Connection 'Internal-LAN' (9d2bb058-c121-4175-8bd3-03d422fad409) successfully ad
ded.
```

```
[root@localhost ~]# nmcli connection modify "wired connection 1" connection.id
nternal-LAN2
[root@localhost ~]# nmcli connection show
NAME           UUID                                  TYPE      DEVICE
enp0s3         1cd79c34-f650-3d92-ae10-8f9710d22b92  ethernet  enp0s3

Internal-LAN   db5678ae-70df-4faa-8935-727d7745ef82  ethernet  --
Internal-LAN2  9d2bb058-c121-4175-8bd3-03d422fad409  ethernet  --
```

```
connection.id:                          Internal-LAN
connection.uuid:                        db5678ae-70df-4faa-8935-727d7745ef82
connection.stable-id:                   --
connection.type:                        802-3-ethernet
connection.interface-name:              enp0s3
connection.autoconnect:                 yes
connection.autoconnect-priority:        0
connection.autoconnect-retries:         -1 (default)
connection.multi-connect:               0 (default)
connection.auth-retries:                -1
connection.timestamp:                   0
connection.permissions:                 --
connection.zone:                        --
connection.controller:                  --
connection.master:                      --
connection.slave-type:                  --
connection.port-type:                   --
connection.autoconnect-slaves:          -1 (default)
connection.autoconnect-ports:           -1 (default)
connection.down-on-poweroff:            -1 (default)
connection.secondaries:                 --
connection.gateway-ping-timeout:        0
connection.ip-ping-timeout:             0
lines 1-23
```

```
[root@localhost ~]# nmcli connection modify  Internal-LAN ipv4.method manual ipv
4.addresses 192.168.1.100/24 ipv4.gateway 192.168.1.1 ipv4.dns 8.8.8.8
```

```
ipv4.method:                            manual
ipv4.dns:                               8.8.8.8
ipv4.dns-search:                        --
ipv4.dns-options:                       --
ipv4.dns-priority:                      0
ipv4.addresses:                         192.168.1.100/24
ipv4.gateway:                           192.168.1.1
```

RESULT:

Thus the study of various network commands used in Linux and windows is done and verified

15/8/24 10/10